ETL DESIGN & OPERATIONAL DOCUMENT: WEB VISIT INGESTION PIPELINE

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PROJECT: VINKOS TECHNICAL ASSESSMENT

STACK: PENTAHO DATA INTEGRATION (PDI) 9.4 / POSTGRESQL 15

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1. Executive Summary

The proposed solution implements a robust, automated ETL pipeline to ingest web visit logs from a remote SFTP server into a centralized Data Warehouse. The architecture addresses the critical requirements of **idempotency** (preventing duplicate loads), **data quality enforcement**, and **operational traceability**.

Key technical challenges—such as handling enterprise-level email lengths and strict database type casting—were successfully resolved during the development phase, resulting in a production-ready system capable of processing thousands of records with zero manual intervention.

2. Technical Architecture

The system follows a modular, layered architecture to ensure separation of concerns and future scalability.

A. High-Level Orchestration (The Master Job)

The workflow is managed by `Job_vinkOS.KJB`, which executes the following control points:

1. **Retrieval:** Securely fetches `report_*.txt` files from the remote SFTP directory.
2. **Staging:** Moves files to a local environment to ensure process stability.
3. **Idempotency Gate:** Queries the `control_cargas` table to verify if a file has been processed previously. If detected, the file is skipped.
4. **Transformation:** Triggers the record-level logic.
5. **Archiving:** Compresses processed files into a ZIP backup and clears the source directory.

B. Data Processing Flow (The Transformation)

The core logic resides in `Transformation_vinkOS.KTR`:

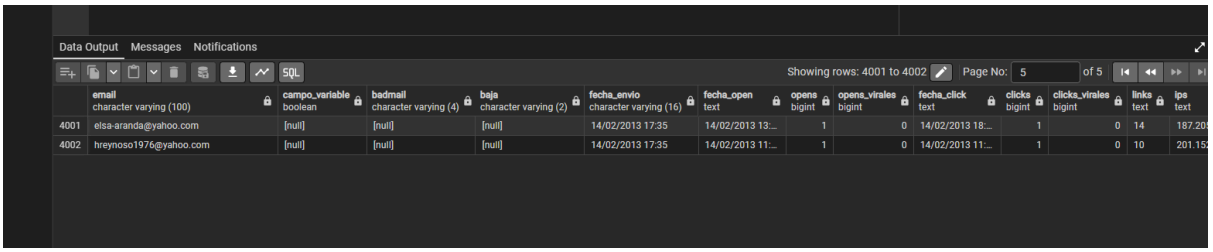
- **Text File Input:** Reads multiple files using wildcard patterns.
- **Metadata Injection:** Explicitly casts incoming strings into **Boolean** and **Integer** types to ensure RDBMS compatibility.
- **Data Quality Gate:** A `Data Validator` step filters records based on email syntax and `dd/MM/yyyy HH:mm` date patterns.

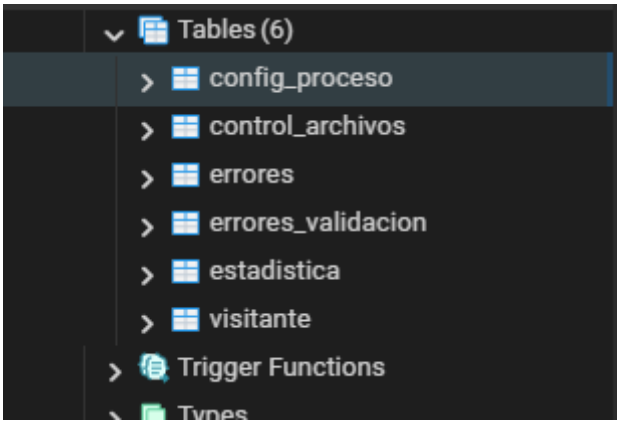
3. Database Schema & Evidence

Schema Hardening

During testing, the schema was optimized for production stability:

- **Email Length:** Standardized to `VARCHAR(100)` to prevent overflows from long enterprise addresses.
- **Snake_Case Sync:** All column names were synchronized (e.g., `fecha_envio`) to match the PDI stream and avoid "Field not found" errors.





4. Operational Results & Monitoring

The pipeline provides full visibility into the ingestion health through a dedicated audit layer.

Execution Metrics

- **Total Records Read:** 2,004 (from 3 source files).
- **Successful Loads:** 2,001 (inserted into `estadistica` and synchronized with `visitante`).
- **Validation Errors:** 3 (redirected to `errores_validacion` for manual audit).

5. Operations & Troubleshooting Guide

For the Operations Team, the following monitoring tools are provided:

- **Monthly Report View:** `SELECT * FROM v_reporte_mensual_etl;` provides a high-level summary of processed files and error rates.
- **Incident Response:** * *Symptom:* Records appearing in `errores_validacion`.
    - *Action:* Review the `desc_error` column. Common causes include malformed email strings or incorrect date formats in the source TXT.

6. Big Data Scalability Roadmap

If daily volume scales to millions of records, the architecture is designed for a seamless transition:

- **Orchestration:** Migration from PDI to **Apache Spark**.
- **Storage:** Transition from local filesystem to **Object Storage (S3 / HDFS)**.
- **Data Lake:** Implementation of **Delta Lake or Hive** tables to replace the relational instance.